

Figure 1: A very simple IVR

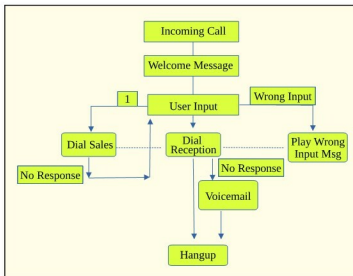


Figure 2: IVR with user input (all details are not shown)

could contain a voice recording of the message, “Welcome to OSFY.” *EXTEN* saves the value of the extension dialled by the caller. Dial the *EXTEN* using the SIP protocol and ring for 60 seconds. The user may pick up the call and talk to the caller. If the user is unavailable, call the service’s voicemail with the same *EXTEN* extension number. After returning from the voicemail, hang up.

Example 2

The next example demonstrates how calls can be routed based on the user’s inputs:

```
[from-pstn]
exten => 1234,1,Answer();
exten => 1234,n,Set(TIMEOUT(digit)=1);
exten => 1234,n,Set(TIMEOUT(response)=10);
exten => 1234,n,Background(welcome);
exten => 1234,n,Background(ivr-options);
exten => 1234,n,WaitExten();
```

The welcome message is played in the background, if the user dials the extension 1234. The function *Playback()* is blocked and the user will be able to provide the inputs only after the message is completed. In case of *Background*, the welcome message and *ivr-options* are played one after the other. The users can input their choice at any point of time. The function *TIMEOUT* is set for two cases: 1) if the user presses one digit, and 2) if the time exceeds 10 seconds. Also, note that the second parameter ‘n’ takes away the burden of sequencing, like in Example 1, and makes the sequence dynamically next to the previous statement. The *ivr-options* plays the message, “Please press 1 for sales, 2 for support, 3 for operator...”

```
exten => 1,1,Dial(SIP/2000&SIP/2001);
exten => 1,n,Playback(sendback-to-ivr);
exten => 1,n,Goto(1234,1);
exten => 2,1,Dial(SIP/2002&SIP/2003);
exten => 2,n,Playback(sendback-to-ivr);
exten => 2,n,Goto(1234,1);
```

If the user presses 1, the extensions 2001 and 2002 will ring in parallel. If no one picks up, a voice file stating that, “Currently, no agents are available,” is played and the call is sent back to the main IVR loop. Similarly, if the user presses 2, both the extensions 2003 and 2004 in the sales department will ring.

```
exten => 0,1,Dial(SIP/2111,50);
exten => 0,n,Voicemail(2111,u);
exten => 0,n,Hangup();
```

If the user presses 0 to talk to the operator, the extension 2111 will ring for 50 seconds. If nobody responds, then the call is redirected to the voice mail.

```
exten => 1,1,NOOP(wrong input received);
exten => 1,n,Playback(Invalid);
exten => 1,n,Goto(1234,1);
```

If the user presses anything other than 1, 2 or 0, the message file with, “You have chosen an invalid input,” is played and the call is sent back to the main loop.

```
exten => t,1,NOOP(no input received);
exten => t,n,Playback(pls-select-option);
exten => t,n,Goto(1234,1);
```

If the user comes out of the loop without any input due to the timeout setting of 10 seconds, then another message, “You have not selected any input,” is played and sent back to the main loop.

The dial plan also provides the choice to query and store to an external database. In the next example, we will have